

UrbanMind: A Unified Explainable GenAI Framework for Anticipatory, Risk-Aware Urban Multimodal Mobility Management

Nikhil Kumar¹, Ruchi Mishra², Lakshya Sharma³, Aayush Mathur³,
Dr. Nandan Maiti^{4*}



¹IIT Kottayam, ²Banasthali Vidyapeeth
³Rajasthan Technical University
⁴Indian Institute of Technology Kharagpur
*nandan@civil.iitkgp.ac.in

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Introduction

Urban mobility in Indian cities is highly dynamic, disrupted by weather, public events, infrastructure works, and socioeconomic activities. While digital mapping and traffic tools are widespread, most remain reactive - relying on structured traffic records and transit schedules. They often miss contextual signals from traffic APIs, social media, public notices, and weather alerts, causing less reliable routing decisions in rapidly changing urban environments.

Moving from reactive navigation to proactive disruption-aware mobility.

Problem Statement

Pain points:

Reactive Routing

Current systems respond after disruptions occur, rather than predicting mobility risks in advance.

Unused Real-Time Signals

Valuable information from traffic APIs, news, weather alerts, and social sources remains under-utilised.

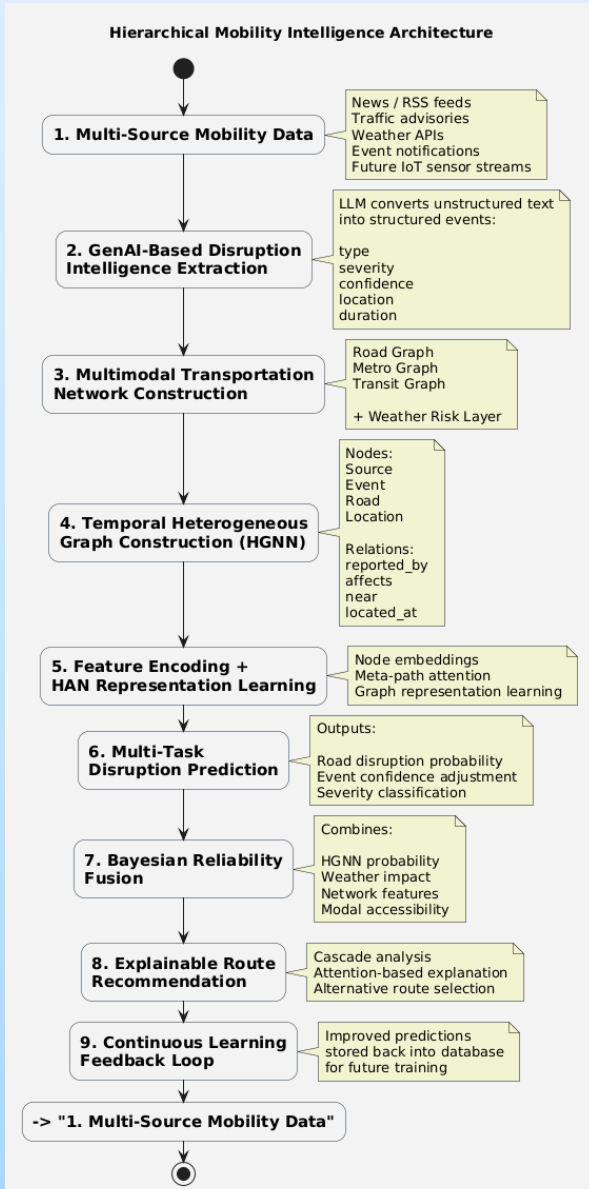
Limited Explainability

Route decisions often lack transparency, reducing user trust in intelligent mobility systems.

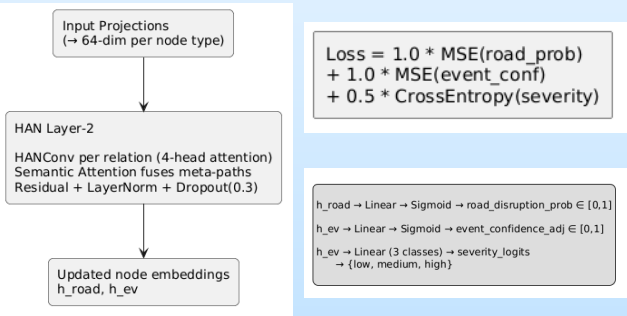
How can real-time urban data be converted into explainable intelligence for proactive and personalized mobility decisions?

System Architecture

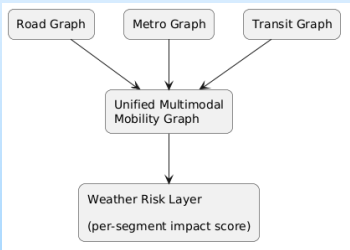
UrbanMind integrates heterogeneous mobility data with GenAI extraction, HGNN-based spatial reasoning, Bayesian risk fusion, and explainable route optimization to enable proactive multimodal travel decisions.



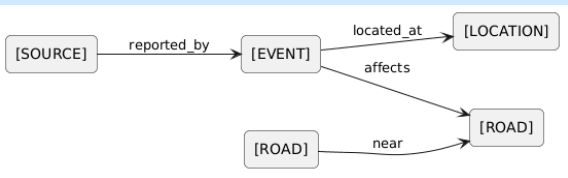
Heterogeneous Attention Network & Bayesian Reliability Fusion



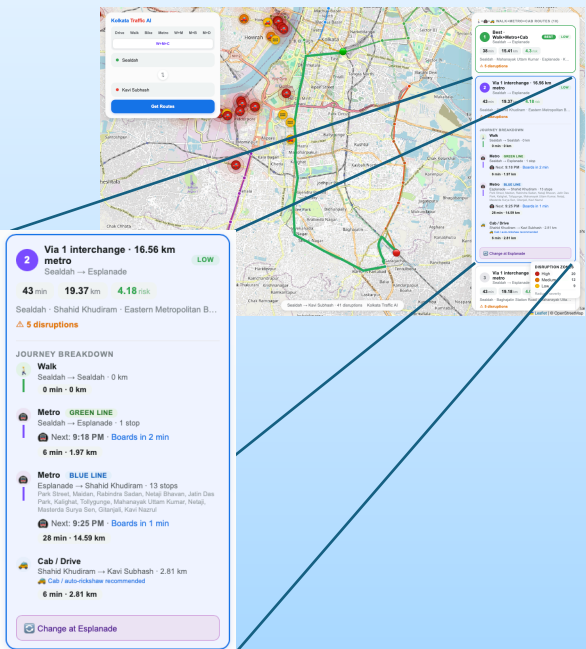
Multimodal Routing Engine



Temporal Heterogeneous Graph Construction (HGNN)



Prototype Demonstration



Positioning Against Existing Mapping Services

Aspect	Existing Systems	UrbanMind
Data Sources	Structured + Crowd	Structured + Unstructured
Event Handling	Reactive Updates	Future-aware Prediction
Uncertainty	Fixed Decisions	Bayesian Risk Score
Route Selection	Shortest Time	Risk + Reliability
Explainability	Limited	HGNN Attention
Mobility Modes	Limited	Multimodal Routing

Acknowledgement & References

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